

전장

작업 지도서

Smart-IC AVI 2 ~ 4호기

(LG 이노텍)

2016.08.25

작업 지도서 1

[FERRITE CORE (페라이트 코어) 장착 (16.04.23)]

- 1) SERVO AMP 전면 케이블에 페라이트 코어를 장착 한다.
 1. SERVO AMP Axis No. : Axis 0 ~ Axis 7
 2. CN1 (Commander Cable) : ZCAT3035-1330A (내경 13mm)
 3. CN2 (Encoder Cable) : ZCAT2035-0930A (내경 9mm)
- 2) 참고자료 : 사진 1

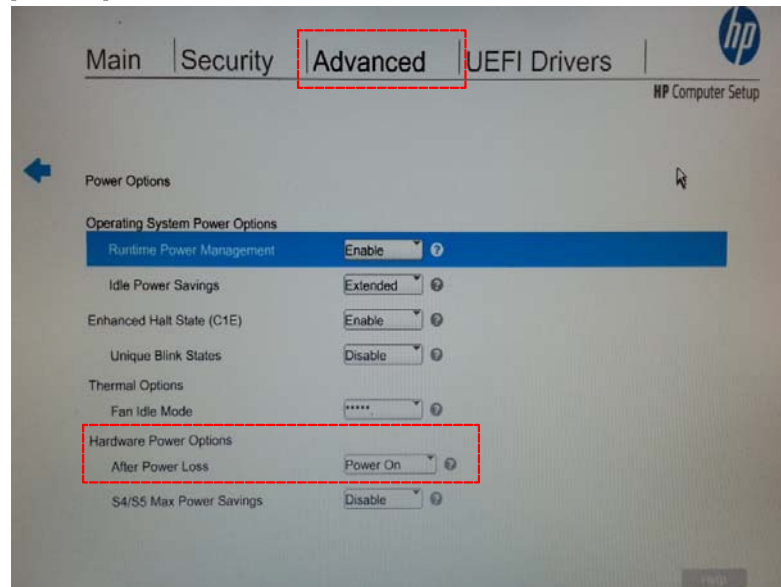
[사진 1]



[VISION PC CMOS SETTING (16.04.25)]

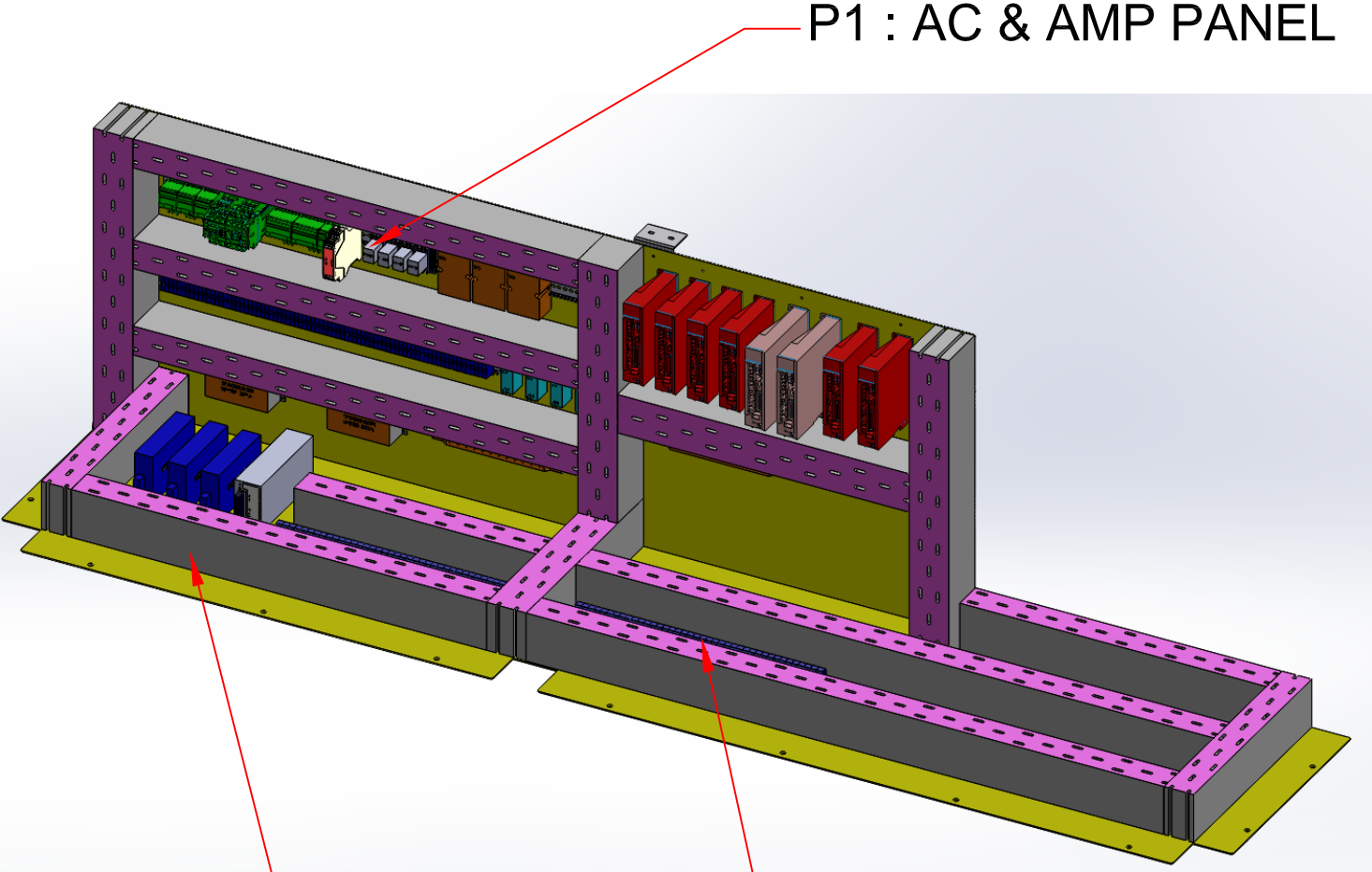
- 1) Setting 항목.
 1. Hardware Power Option → After Power Loss : Power On.
- 2) Setting 목적.
 1. UPS1 Power On 하면 Vision PC 6대를 자동으로 부팅 하기 위함.
- 3) Setting 방법.
 1. PC Turn On → F10 버튼 누름. (모니터에 HP 로고가 보일때 누른다)
 2. CMOS 진입
 3. [Advanced] → Power Option → Hardware Power Option → After Power Loss
 4. After Power Loss : Power On 으로 설정 한다.
 5. [Main] → Save Changes and Exit → Yes.
 6. 자동으로 Re-Booting 된다.
- 4) 적용
 1. VISION PC 1-1 ~ 3-2 (6대)
- 5) 참고 자료 : 사진 2

[사진 2]



Designed	Modify	Approved	Material	Quantity	Treatment	Remarks
김관호 조득희		김관호	Machine name : SMART-IC AVI			
Date:		Unit : mm		Part name : 작업 지도서 1		
Synapse imaging			Size A3	Drawing No.		Sheet No G1

CONTROL PANEL LAYOUT



P1 : AC & AMP PANEL

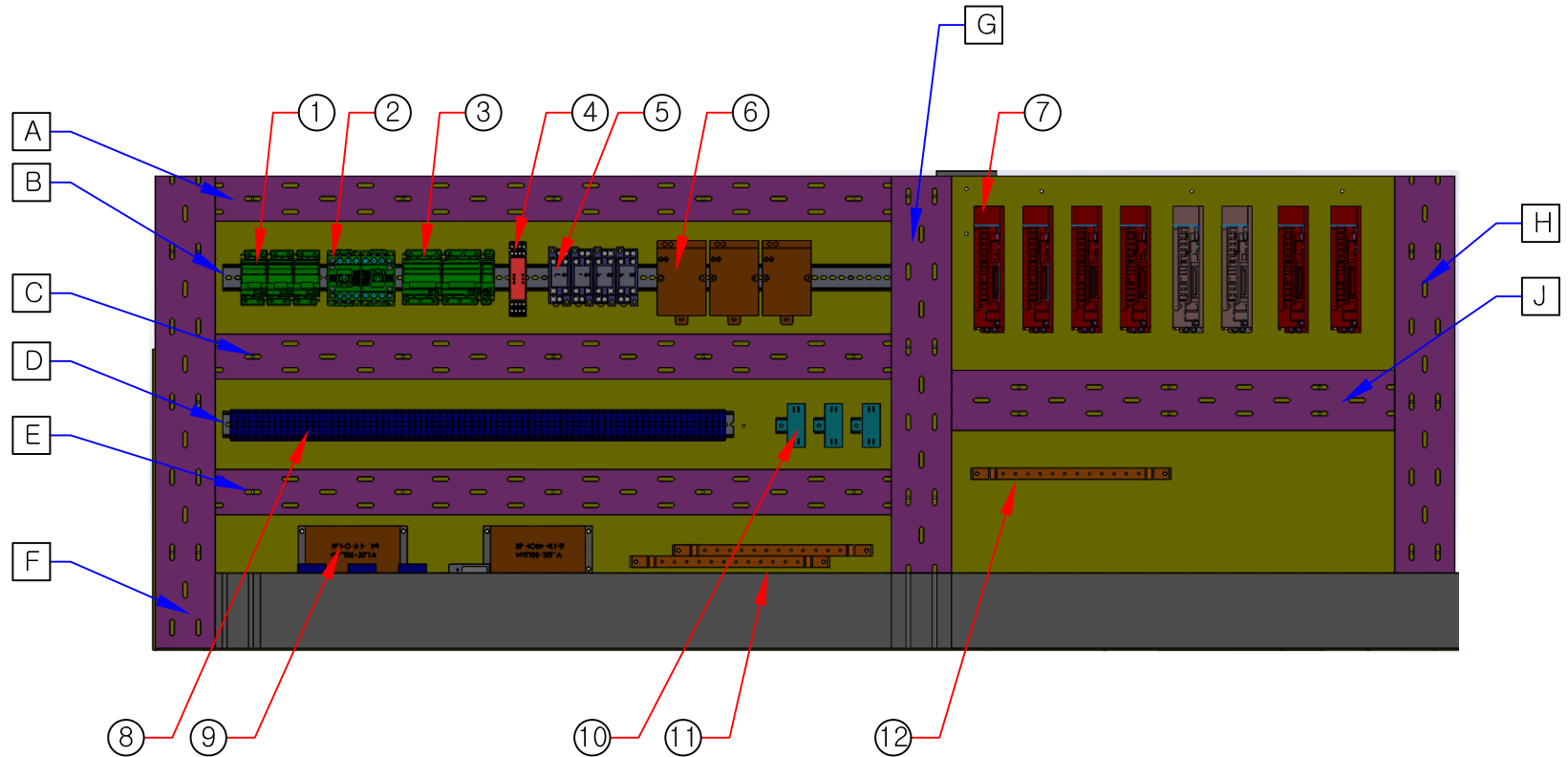
P2 : DC & SIGNAL PANEL(2)

P2 : DC & SIGNAL PANEL(1)

Designed	Modify	Approved	Material	Quantity	Treatment	Remarks
김관호		김관호	Machine name : SMART-IC AVI			
Date: 16.08.25		Unit : mm		Part name : CONTROL PANEL LAYOUT		
Synapse imaging			Size	Drawing No.		Sheet No
			A3			L1

P1 : AC & AMP PANEL

NO	LABEL	PARTS NAME	수량
①	CP 1	GCP-32ANM10A	1
	CP 2	GCP-32ANM10A	1
	CP 3	GCP-32ANM10A	1
②	MC 1	MC-12b-DC24V	1
	MC 2	MC-12b-DC24V	1
③	CP 4	GCP-33ANM10A	1
	CP 5	GCP-33ANM10A	1
	CP 11	GCP-31ANM10A	1
④	SC 1	G9SX-BC202RT	1
⑤	RY 1/3/4/5	LY2N-D2-24VDC	4
		PTF08A	4
⑥	AXIS 10	TMP-1	1
	AXIS 12	TMP-1	1
	AXIS 13	TMP-1	1
			1
⑦	AXIS 7	MR-J4-20A	1
	AXIS 6	MR-J4-20A	1
	AXIS 5	MR-J4-20A	1
	AXIS 4	MR-J4-20A	1
	AXIS 3	MR-J4-40A	1
	AXIS 2	MR-J4-40A	1
	AXIS 1	MR-J4-20A	1
	AXIS 0	MR-J4-20A	1
⑧	AC.TB 3L / 5L	KTB2-015,15A	78
⑨	NF 4 ~ 5	WYFT10T1A	1
⑩	CONDENSER 1/3/4		3
		CH40BFAUL	1
⑪	E3 BAR 1 ~ 2	Cu/367*15*3/M4*12P/고정홀 M6*2	2
⑫	E3 BAR 3	Cu/367*15*3/M4*12P/고정홀 M6*2	1
⑬	RY 5	LY2N-D2-24VDC	1
		PTF08A	1
A	PVC DUCT	60 x 100	1
B	DIN RAIL	KTB2-CHB-2m	1
C	PVC DUCT	60 x 100	1
D	DIN RAIL	KTB2-CHB-2m	1
E	PVC DUCT	60 x 100	1
F	PVC DUCT	80 x 100	1
G	PVC DUCT	80 x 100	1
H	PVC DUCT	80 x 100	1
J	PVC DUCT	80 x 100	1



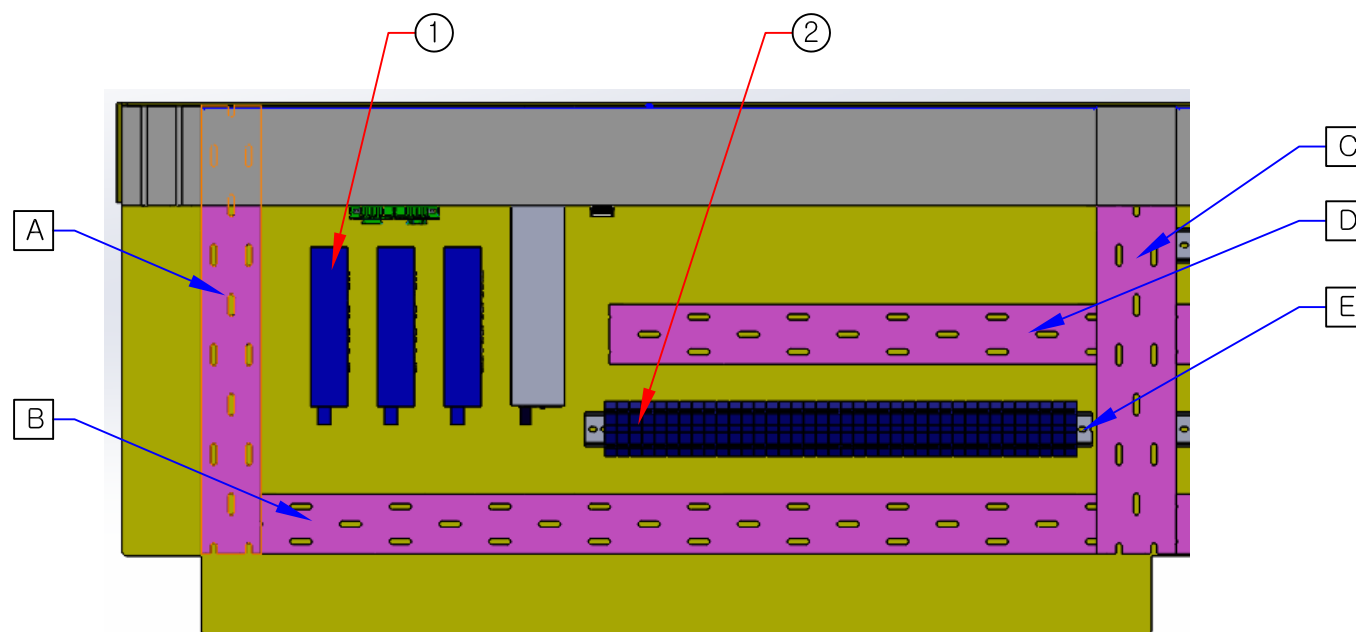
[작업 노트]

★ 2~4호기 수정 내역.

- 삭제 : Axis.11 (종속 부품 모두 삭제)
- Torque Motor Drive (TMP-1) / Relay 2 / Condenser 2
- 사양 변경 : Axis.1
- 변경 전 HA-800A-1C-200
- 변경 후 MR-J4-20A

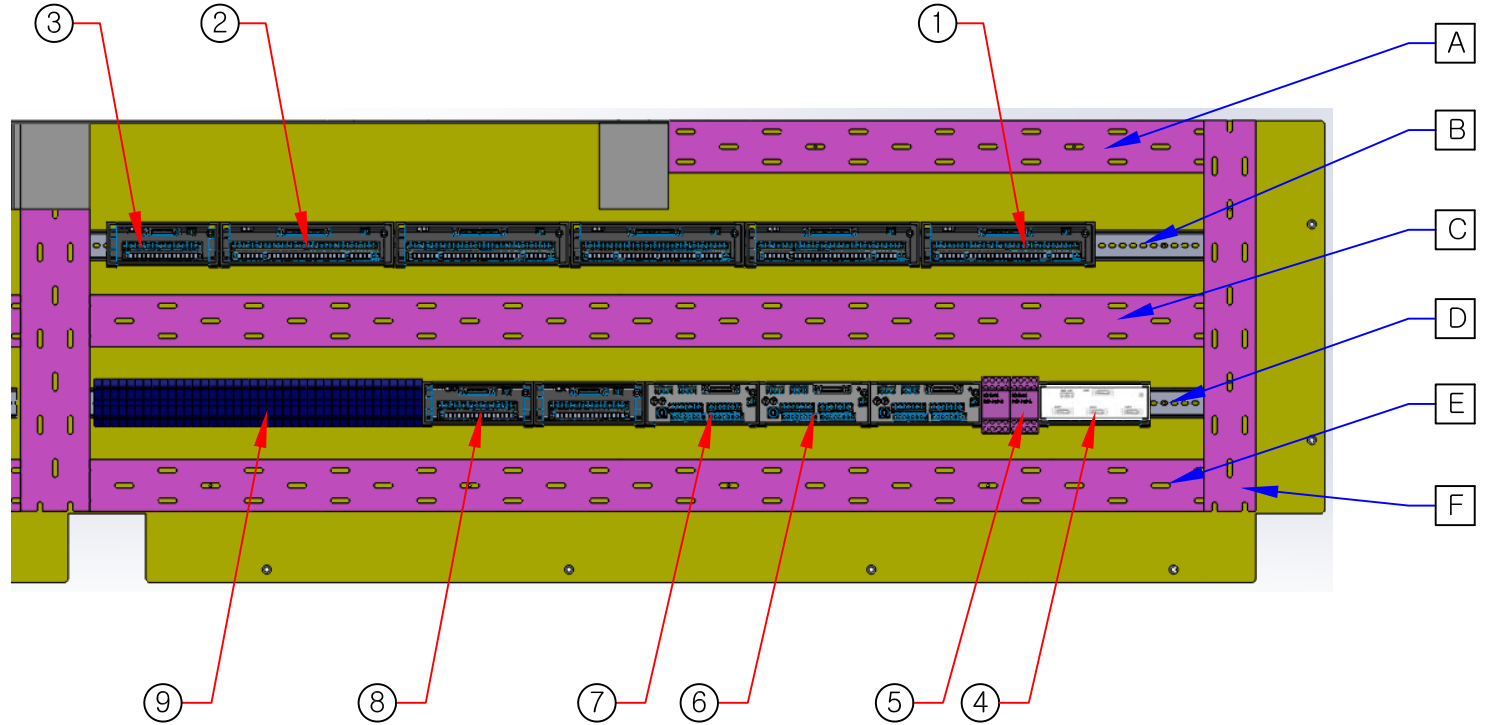
Designed	Modify	Approved	Material	Quantity	Treatment	Remarks
김관호		김관호	Machine name : SMART-IC AVI			
Date: 16.08.25	Unit : mm		Part name : P1 : AC & AMP PANEL			
Synapse imaging		Size A3	Drawing No.		Sheet No L2	

P2 : DC & SIGNAL PANEL (1)

[illegible]

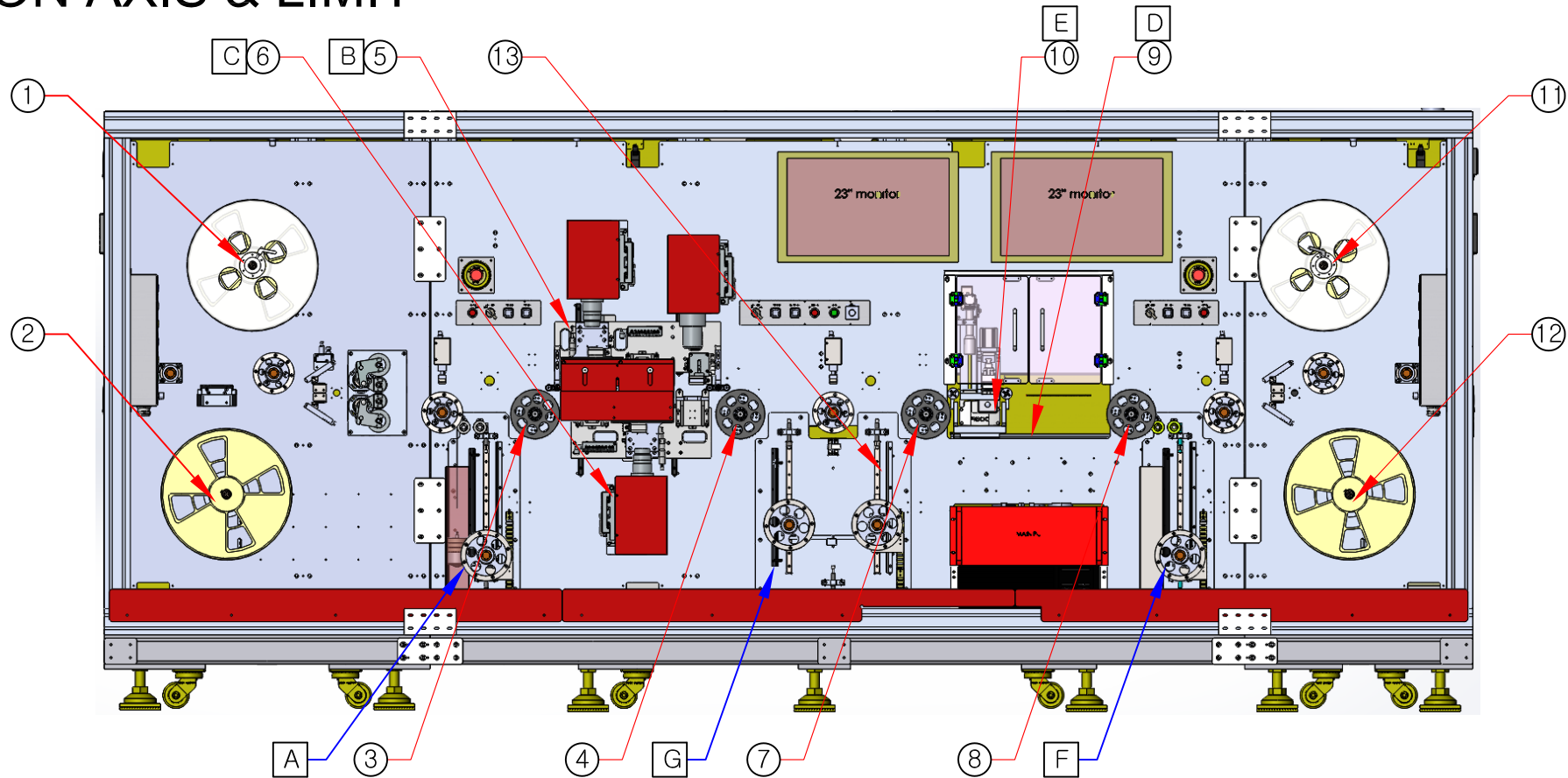
Designed	Modify	Approved	Material	Quantity	Treatment	Remarks
김관호		김관호	Machine name : SMART-IC AVI			
조득희						
Date:		Unit : mm	Part name :			
			P2 : DC & SIGNAL PANEL (1)			
		Size	Drawing No.			Sheet No
		A3				L3

P2 : DC & SIGNAL PANEL (2)

[illegible]

Designed	Modify	Approved	Material	Quantity	Treatment	Remarks
김관호 조득희		김관호	Machine name : SMART-IC AVI			
Date: 16.08.25		Unit : mm	Part name : P2 : DC & SIGNAL PANEL (2)			
			Size A3	Drawing No.		Sheet No L4

MOTION AXIS & LIMIT



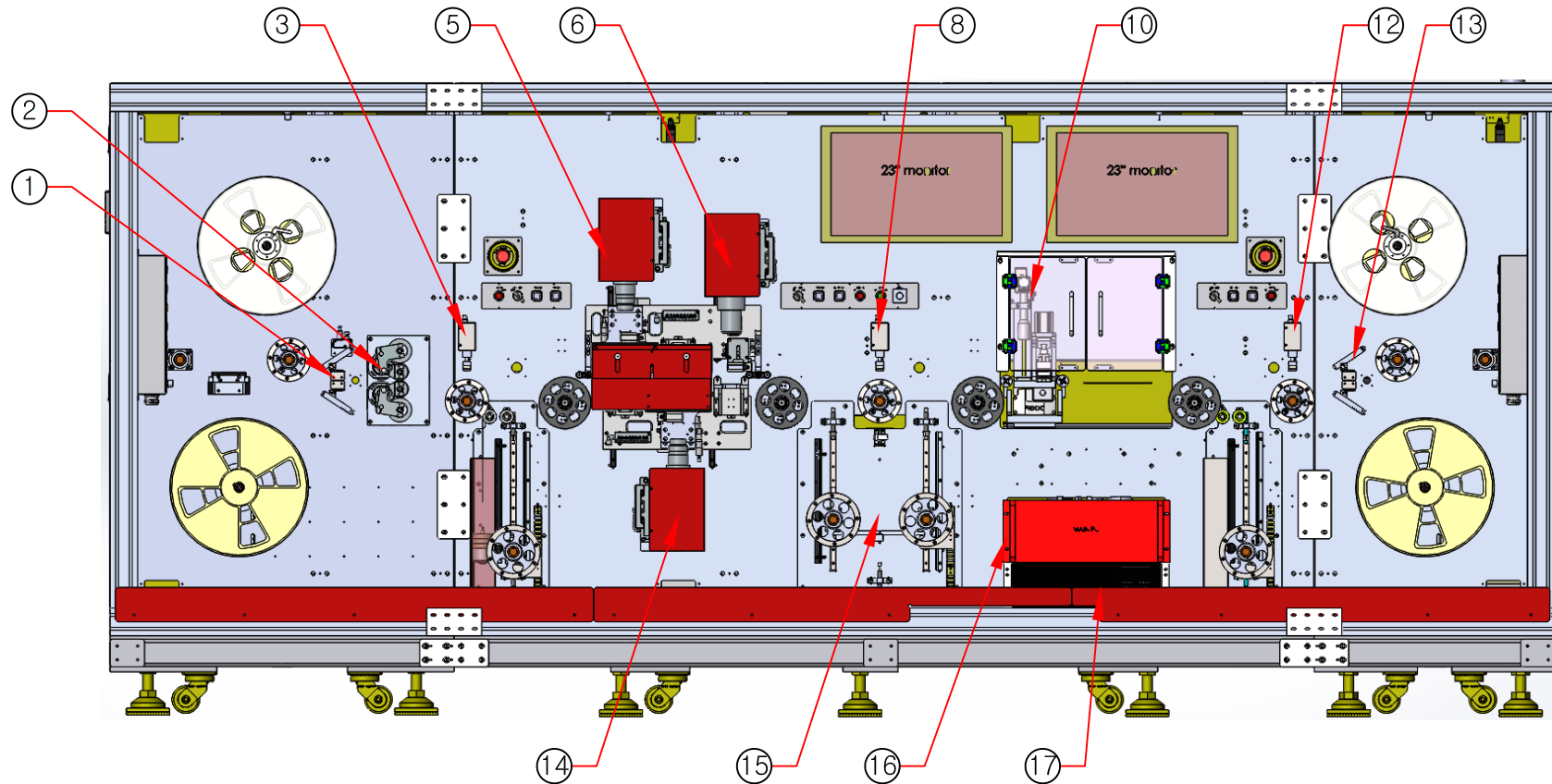
NO	LABEL	AXIS NAME	MOTOR TYPE	MOTOR MODEL
①	AXIS 0	Reel Uncoiler S-IC	SERVO	HG-KR23
②	AXIS 10	Reel Uncoiler 간지	TORQUE	TM520C-18SE
③		Vision - Uncoiler Sprocket		
④	AXIS 1	Vision - Recoiler Sprocket	SERVO	HG-KR23
⑤	AXIS 2	Vision - Top Z	SERVO	HG-KR43B
⑥	AXIS 3	Vision - Bottom Z	SERVO	HG-KR43B
⑦	AXIS 12	Punch - Uncoiler Sprocket	TORQUE	TM520C-AE
⑧	AXIS 4	Punch - Recoiler Sprocket	SERVO	HG-KR23
⑨	AXIS 5	Align & Punch X	SERVO	HG-KR23
⑩	AXIS 6	Align & Punch Y	SERVO	HG-KR23
⑪	AXIS 7	Reel Recoiler S-IC	SERVO	HG-KR23
⑫	AXIS 13	Reel Recoiler 간지	TORQUE	TM520C-18SE
⑬	AXIS 8	Buffer Roller Position	LINEAR ENCODER	MTE-20

NO	LABEL	LIMIT SENSOR
A	0-CW / CCW	EE-SX672
B	2-CW / CCW	EE-SX672A
C	3-CW / CCW	EE-SX672A
D	5-CW / CCW	D-F9N
E	6-CW / CCW	D-F9N
F	7-CW / CCW	EE-SX672
G	8-CW / CCW	EE-SX672

***Upper = CW.**

Designed	Modify	Approved	Material	Quantity	Treatment	Remarks
김관호		김관호	Machine name : SMART-IC AVI			
Date: 16.08.25	Unit : mm		Part name : MOTION AXIS & LIMIT			
Synapse imaging		Size A3	Drawing No.		Sheet No L11	

MODULE

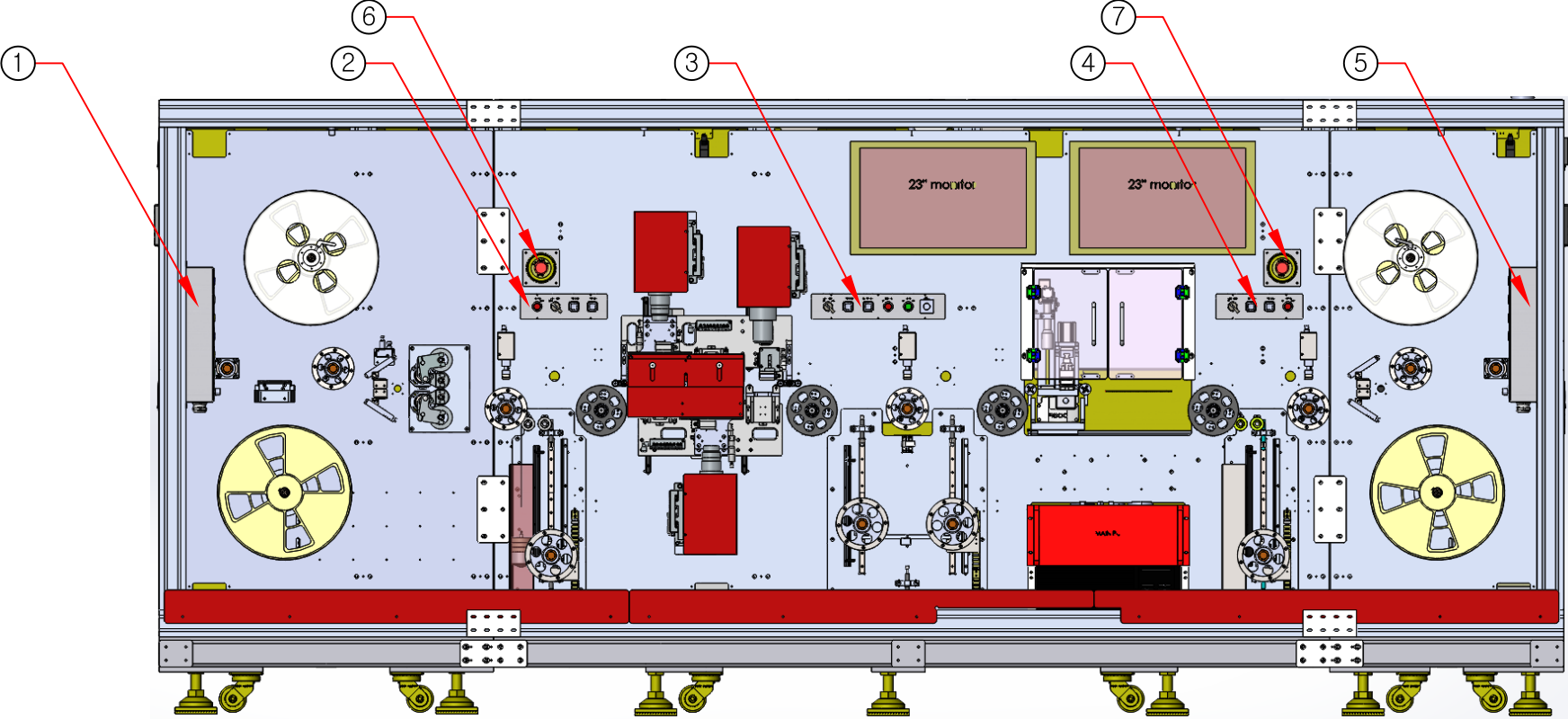


NO	MODULE NAME
①	Loader Ionizer
②	Clean Unit Roller
③	Vision Stopper
⑤	TOP 1 VISION
⑥	TOP 2 VISION
⑧	Buffer Stopper
⑩	Punch Module
⑫	Punch Stopper
⑬	Unloader Ionizer

NO	MODULE NAME
⑭	BOTTOM VISION
⑮	Dual Buffer
⑯	Handler PC
⑰	UPS 1

Designed	Modify	Approved	Material	Quantity	Treatment	Remarks
김관호 조득희		김관호	Machine name : SMART-IC AVI			
Date: 16.08.25		Unit : mm		Part name : MODULE		
Synapse imaging		Size A3		Drawing No.		Sheet No L12

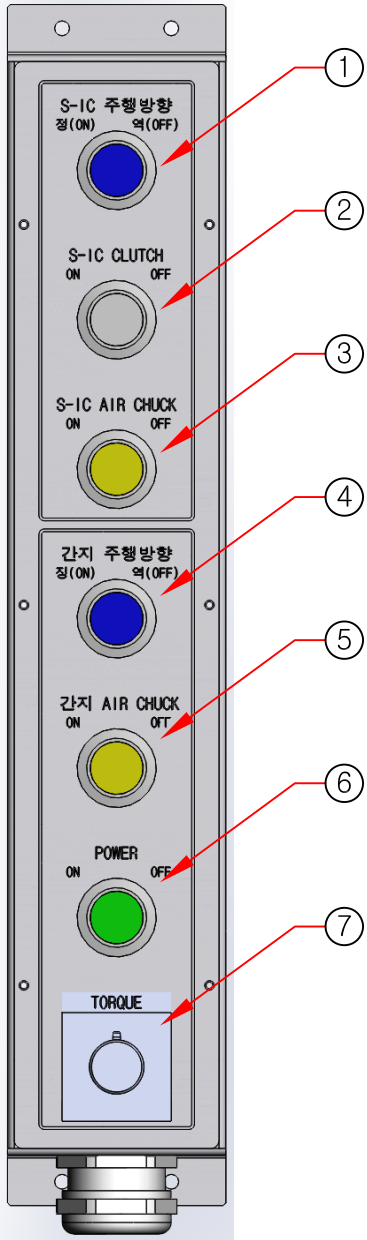
OPERATION SWITCH PANEL LAYOUT



NO	OPERATION SWITCH PANEL NAME
①	LOADER OP
②	VISION OP
③	BUFFER OP
④	PUNCH OP
⑤	UNLOADER OP
⑥	EMO 1 - LEFT
⑦	EMO 2 - RIGHT

Designed	Modify	Approved	Material	Quantity	Treatment	Remarks
김관호 조득희		김관호	Machine name : SMART-IC AVI			
Date: 16.08.25		Unit : mm		Part name : OPERATION SWITCH PANEL LAYOUT		
Synapse  imaging			Size A3	Drawing No.		Sheet No L13

LOADER / UNLOADER OPERATION SWITCH PANEL



[LOADER OP]

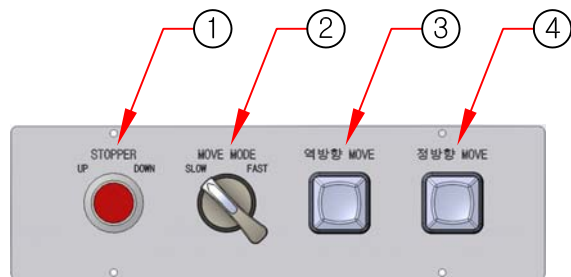
NO	TYPE	COLOR	MODEL	FUNCTION
①	조광형 Push Button	BLUE	KGX-JAD21B	Reel Uncoiler S-IC 구동 방향 설정.
②	조광형 Push Button	WHITE	KGX-JAD21W	Reel Uncoiler S-IC 클러치 On/Off
③	조광형 Push Button	YELLOW	KGX-JAD21Y	Reel Uncoiler S-IC 에어척 Lock/Unlock
④	조광형 Push Button	BLUE	KGX-JAD21B	Reel Uncoiler 간지 구동 방향 설정.
⑤	조광형 Push Button	YELLOW	KGX-JAD21Y	Reel Uncoiler 간지 에어척 Lock/Unlock
⑥	조광형 Push Button	GREEN	KGX-JAD21G	Reel Uncoiler 간지 Torque Motor Drive Power On/Off
⑦	Volume	Blue	PAVR-20KZ	Reel Uncoiler 간지 Torque Motor Control

[UNLOADER OP]

NO	TYPE	COLOR	MODEL	FUNCTION
①	조광형 Push Button	BLUE	KGX-JAD21B	Reel Recoiler S-IC 구동 방향 설정.
②	조광형 Push Button	WHITE	KGX-JAD21W	Reel Recoiler S-IC 클러치 On/Off
③	조광형 Push Button	YELLOW	KGX-JAD21Y	Reel Recoiler S-IC 에어척 Lock/Unlock
④	조광형 Push Button	BLUE	KGX-JAD21B	Reel Recoiler 간지 구동 방향 설정.
⑤	조광형 Push Button	YELLOW	KGX-JAD21Y	Reel Recoiler 간지 에어척 Lock/Unlock
⑥	조광형 Push Button	GREEN	KGX-JAD21G	Reel Recoiler 간지 Torque Motor Drive Power On/Off
⑦	Volume	Blue	PAVR-20KZ	Reel Recoiler 간지 Torque Motor Control

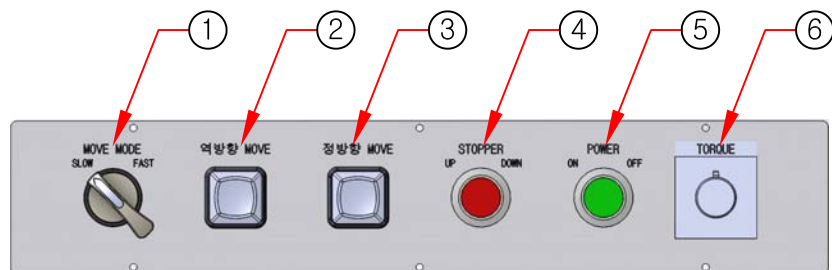
Designed	Modify	Approved	Material	Quantity	Treatment	Remarks
김관호 조득희		김관호	Machine name : SMART-IC AVI			
Date: 16.08.25		Unit : mm		Part name : LOADER / UNLOADER OPERATION SWITCH PANEL		
Synapseimaging			Size A3	Drawing No.		Sheet No L14

VISION / BUFFER / PUNCH OPERATION SWITCH PANEL



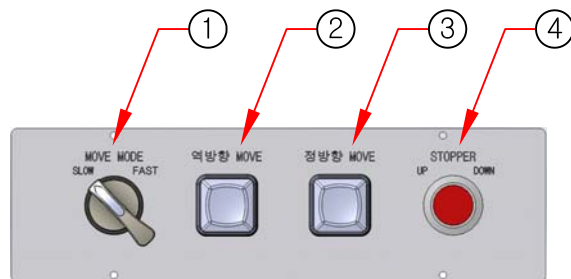
[VISION OP]

NO	TYPE	COLOR	MODEL	FUNCTION
①	조광형 Push Button	RED	KGX-JAD21R	Vision - Stopper Up/Down
②	2단 Select		S2SR-S4WAB	Vision - S-IC Moving Mode Select (SLOW / FAST) 2
③	조광형 Push Button	BLUE	S2PRS-P3BLAD	Vision - S-IC 역방향 Moving 2
④	조광형 Push Button	BLUE	S2PRS-P3BLAD	Vision - S-IC 정방향 Moving 2



[BUFFER OP]

NO	TYPE	COLOR	MODEL	FUNCTION
①	2단 Select		S2SR-S4WAB	Vision - S-IC Moving Mode Select (SLOW / FAST) 1
②	조광형 Push Button	BLUE	S2PRS-P3BLAD	Vision - S-IC 역방향 Moving 1
③	조광형 Push Button	BLUE	S2PRS-P3BLAD	Vision - S-IC 정방향 Moving 1
④	조광형 Push Button	RED	KGX-JAD21R	Buffer - Stopper Up/Down
⑤	조광형 Push Button	GREEN	KGX-JAD21G	Punch - Uncoiler Sprocke Torque Motor Drive Power On/Off
⑥	Volume		PAVR-20KZ	Punch - Uncoiler Sprocke Torque Motor Control

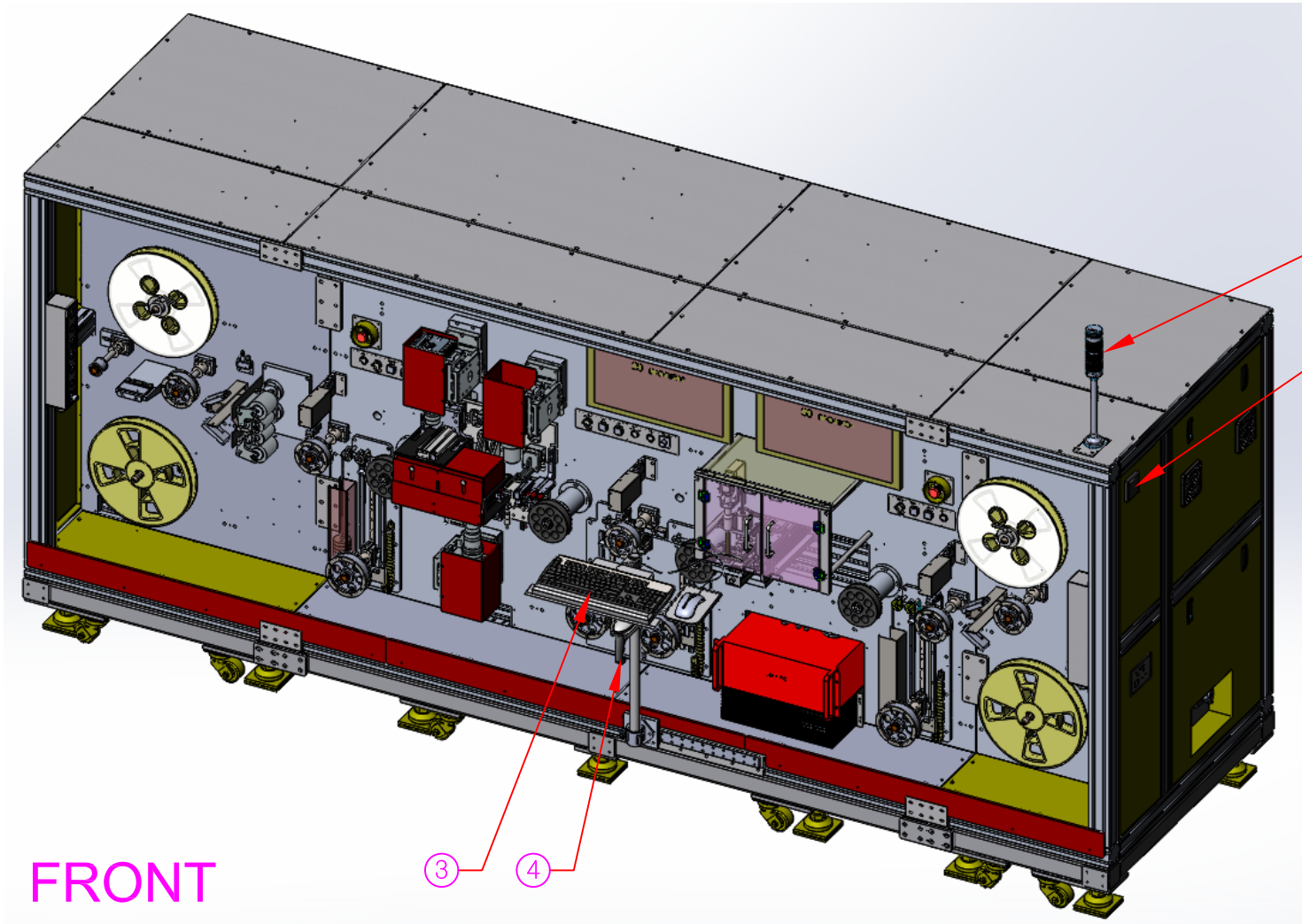


[PUNCH OP]

NO	TYPE	COLOR	MODEL	FUNCTION
①	2단 Select		S2SR-S4WAB	Punch - S-IC Moving Mode Select (SLOW / FAST)
②	조광형 Push Button	BLUE	S2PRS-P3BLAD	Punch - S-IC 역방향 Moving
③	조광형 Push Button	BLUE	S2PRS-P3BLAD	Punch - S-IC 정방향 Moving
④	조광형 Push Button	RED	KGX-JAD21R	Punch - Stopper Up/Down

Designed	Modify	Approved	Material	Quantity	Treatment	Remarks
김관호 조득희		김관호	Machine name : SMART-IC AVI			
Date: 16.08.25		Unit : mm		Part name : VISION / BUFFER / PUNCH OPERATION SWITCH PANEL		
Synapse imaging		Size A3		Drawing No.		Sheet No L15

ISOMETRIC VIEW



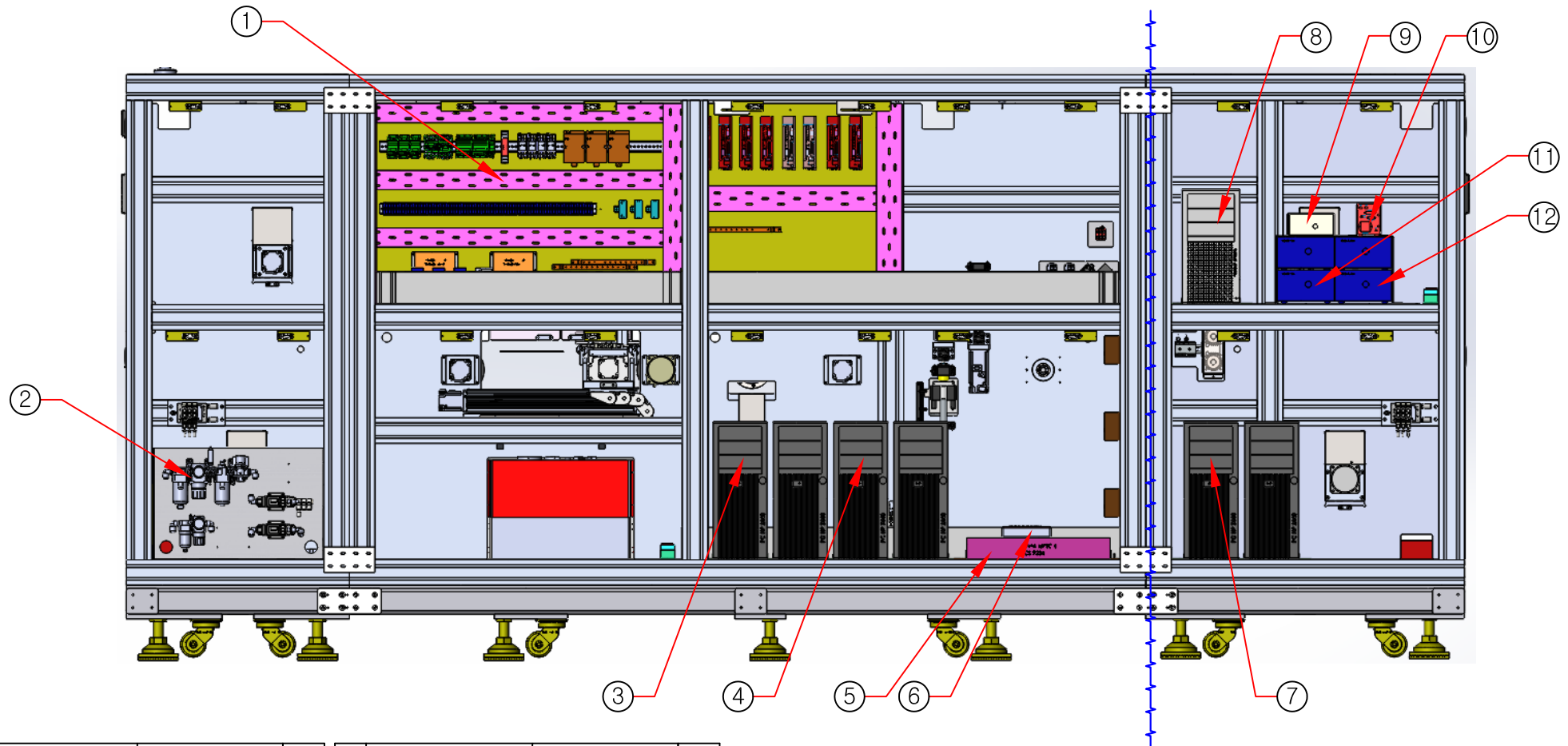
FRONT

NO	LABEL	NAME	수량
①	TOWER LAMP	PTE-DP-302-RYG	1
②	BUZZER	KSP-42S	1
③	KEYBOARD	104Key, LED, USB	1
	MOUSE	Optical Wheel, USB	1

NO	LABEL	NAME	수량
④	BARCODE READER	LS-2208AP	1
		USB (코일형 USB 케이블)	

Designed	Modify	Approved	Material	Quantity	Treatment	Remarks
김관호 조득희		김관호	Machine name : SMART-IC AVI			
Date: 16.08.25		Unit : mm		Part name : ISOMETRIC VIEW		
Synapse  imaging			Size A3	Drawing No.		Sheet No L21

VISION SYSTEM : REAR SIDE 1 (REAR VIEW)



*출하 시 분리 되는 구간 임.

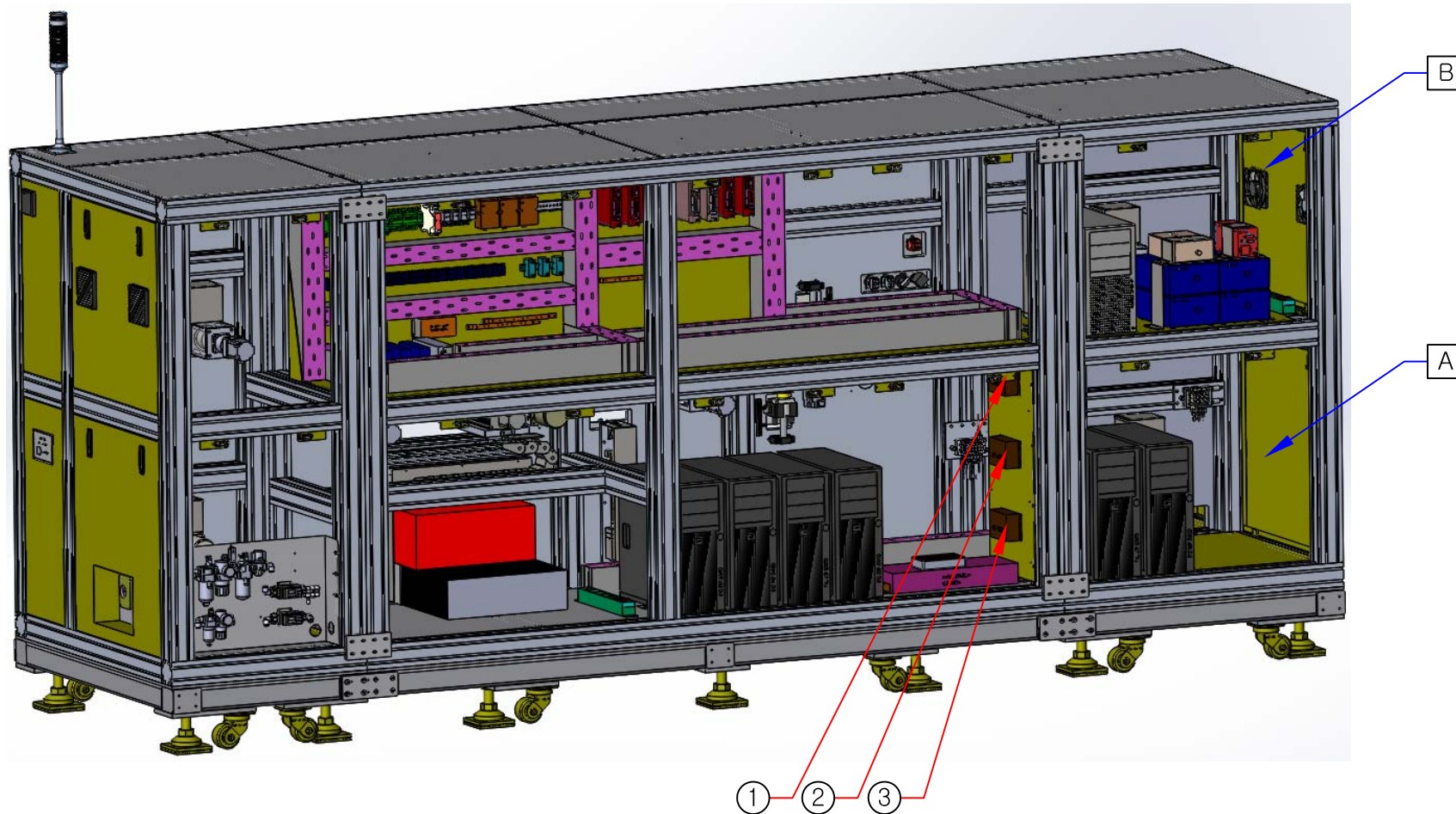
[작업 노트]

1. 출하 시 분리되는 구간이 있으니 전장 배선 작업시 고려 하여 작업 할 것.

NO	LABEL	NAME	수량	NO	LABEL	NAME	수량
①	C-BOX 1	Main Control Panel	1	⑨	LED LIGHT CTRL 3	SPO-LED-2CD-50W	1
②	Main Air Check	ISE30A-01-N-L	1	⑩	LED LIGHT CTRL 6	ALT-2RSD-24V(60W)	1
	Air Flow Switch 1, 2	PF2A721-03-27	2	⑪	LED LIGHT CTRL 1	SPO-LED-4CD-200W	1
③	VISION PC 1-1	HP Z440	1		LED LIGHT CTRL 2	SPO-LED-4CD-200W	1
	VISION PC 1-2	HP Z440	1	⑫	LED LIGHT CTRL 4	SPO-LED-4CD-200W	1
④	VISION PC 2-1	HP Z440	1		LED LIGHT CTRL 5	SPO-LED-4CD-200W	1
	VISION PC 2-2	HP Z440	1				
⑤	KVM SWITCH	CS1758	1				
⑥	L2 SWITCH	HP 1405-8G v2	1				
⑦	VISION PC 3-1	HP Z440	1				
	VISION PC 3-2	HP Z440	1				
⑧	SERVER PC	IBM x3100M5	1				

Designed	Modify	Approved	Material	Quantity	Treatment	Remarks
김관호		김관호	Machine name :	SMART-IC AVI		
조득희			Part name :	VISION SYSTEM : REAR SIDE 1 (REAR VIEW)		
Date: 16.08.25	Unit: mm		Size	Drawing No.	Sheet No	
Synapse imaging			A3		L22	

VISION SYSTEM : REAR SIDE 2 (REAR VIEW)



[작업 노트]

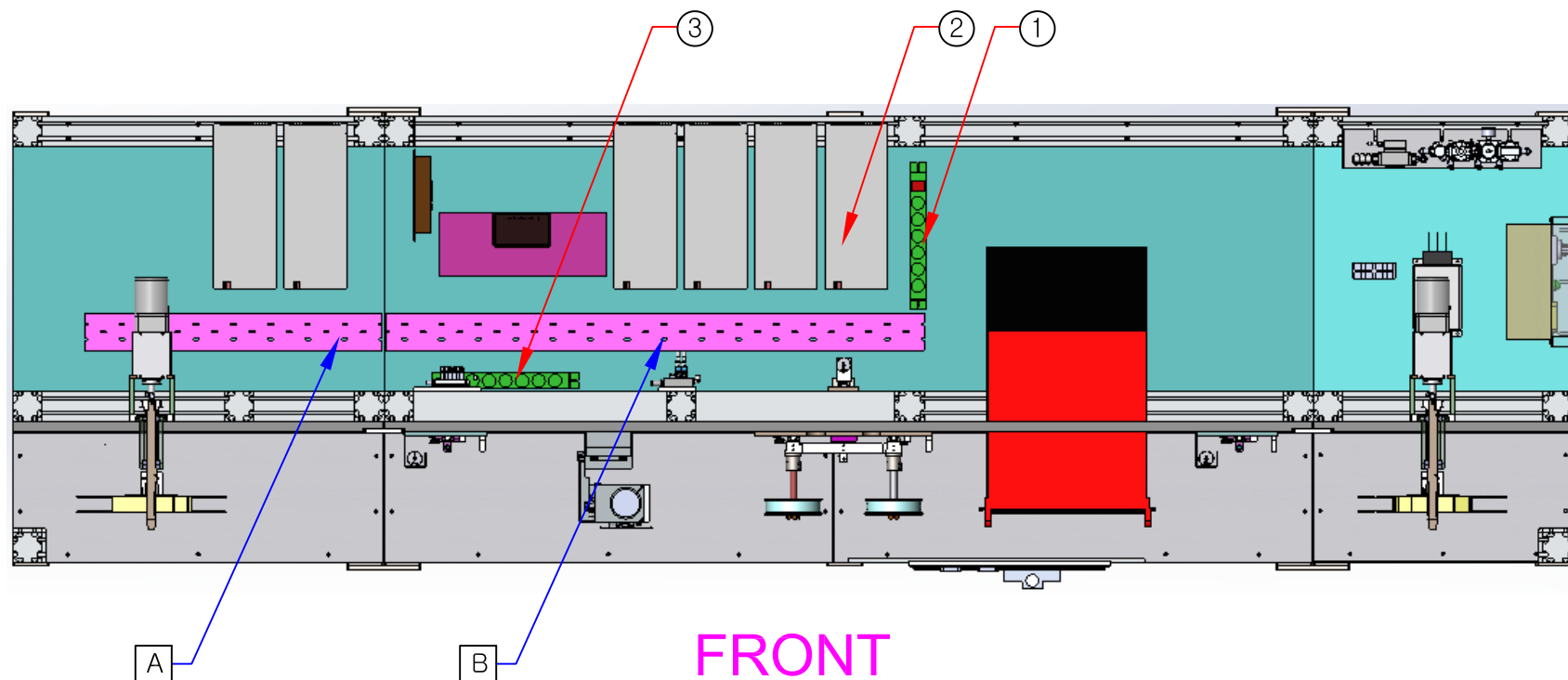
1. Camera Link Cable은 덕트에 매설하지 말고 마운트 및 케이블 타이 사용하여 기중 개방 및 독립 배선 할 것.
(남은 구간은 round 처리해서 Vision PC 3-2 옆의 빈 공간에 정리 한다.)
2. Vision PC 후면에 연결되는 Cable들은 매직 케이블(선정리기)를 사용하여 정리 배선 한다. (후크밴드 또는 헬리컬 밴드 사용 가능)

NO	LABEL	NAME	수량
①	SPLITTER 1	SCL-300T2	1
②	SPLITTER 2	SCL-300T2	1
③	SPLITTER 3	SCL-300T2	1

NO	LABEL	NAME	수량
A	1st FLOOR	Vision System	1
B	2nd FLOOR	Vision System	1

Designed	Modify	Approved	Material	Quantity	Treatment	Remarks
김관호 조득희		김관호	Machine name : SMART-IC AVI			
Date: 16.08.25		Unit : mm	Part name : VISION SYSTEM : REAR SIDE 2 (REAR VIEW)			
Synapse  imaging			Size A3	Drawing No.		Sheet No L23

VISION SYSTEM : 1st FLOOR (TOP VIEW)

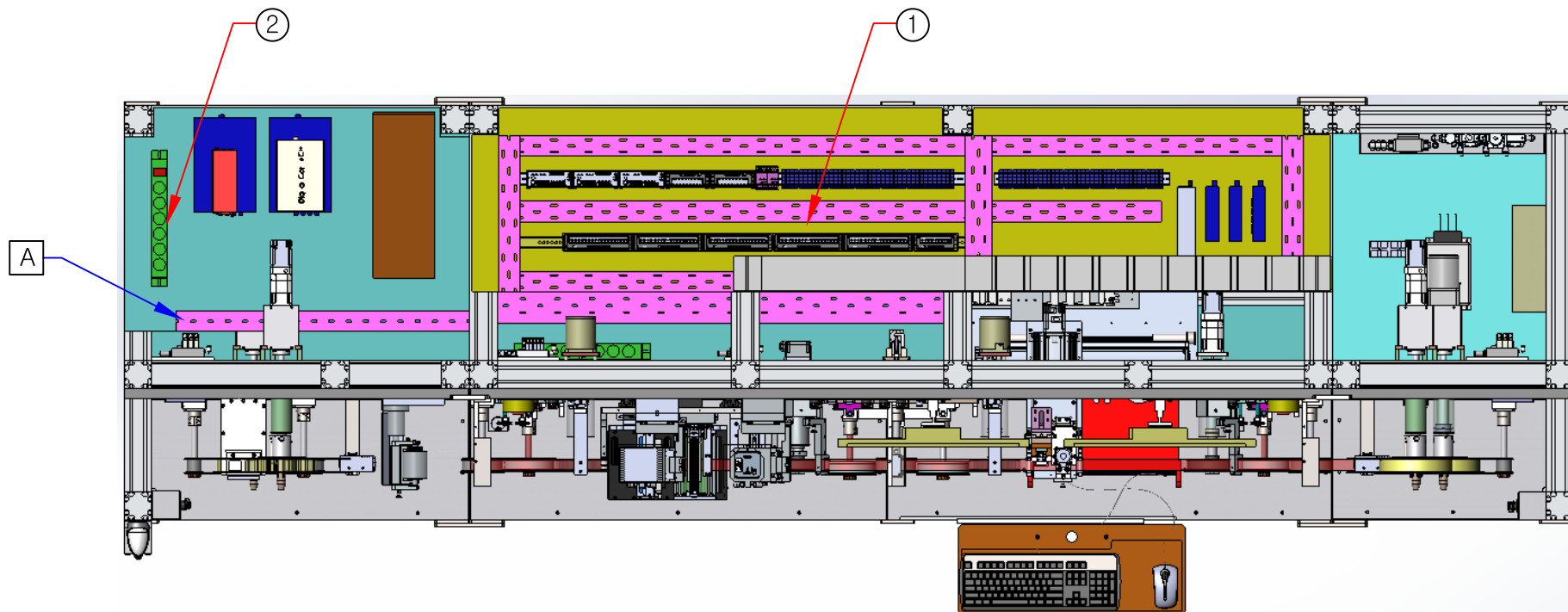


NO	LABEL	NAME	수량
①	멀티탭 1	6구,알루미늄,서버랙	1
②	VISION PC 1-1	HP Z440	1
③	멀티탭 2	6구,알루미늄,서버랙	1

NO	LABEL	NAME	수량
A	PVC DUCT	100 x 100	1
B	PVC DUCT	100 x 100	1

Designed	Modify	Approved	Material	Quantity	Treatment	Remarks
김관호 조득희		김관호	Machine name : SMART-IC AVI			
Date:		Unit : mm	Part name : VISION SYSTEM : 1st FLOOR (TOP VIEW)			
Synapse  imaging			Size A3	Drawing No.		Sheet No. L24

VISION SYSTEM : 2nd FLOOR (TOP VIEW)



FRONT

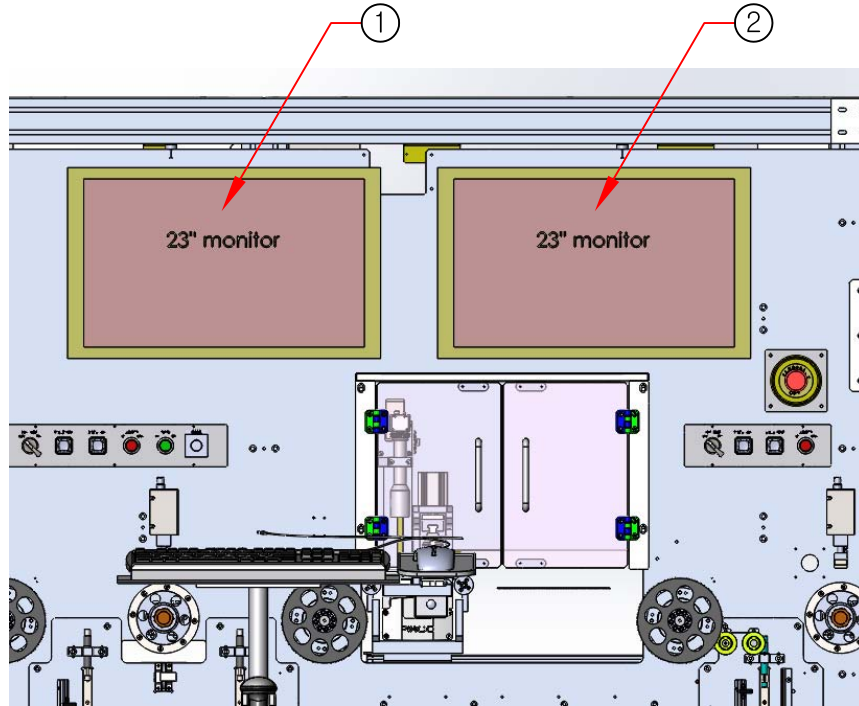
NO	LABEL	NAME	수량
①	C-BOX1	Main Control Panel	1
②	멀티탭 3	6구,알루미늄,서버랙	1

NO	LABEL	NAME	수량
A	PVC DUCT	60 x 80	1

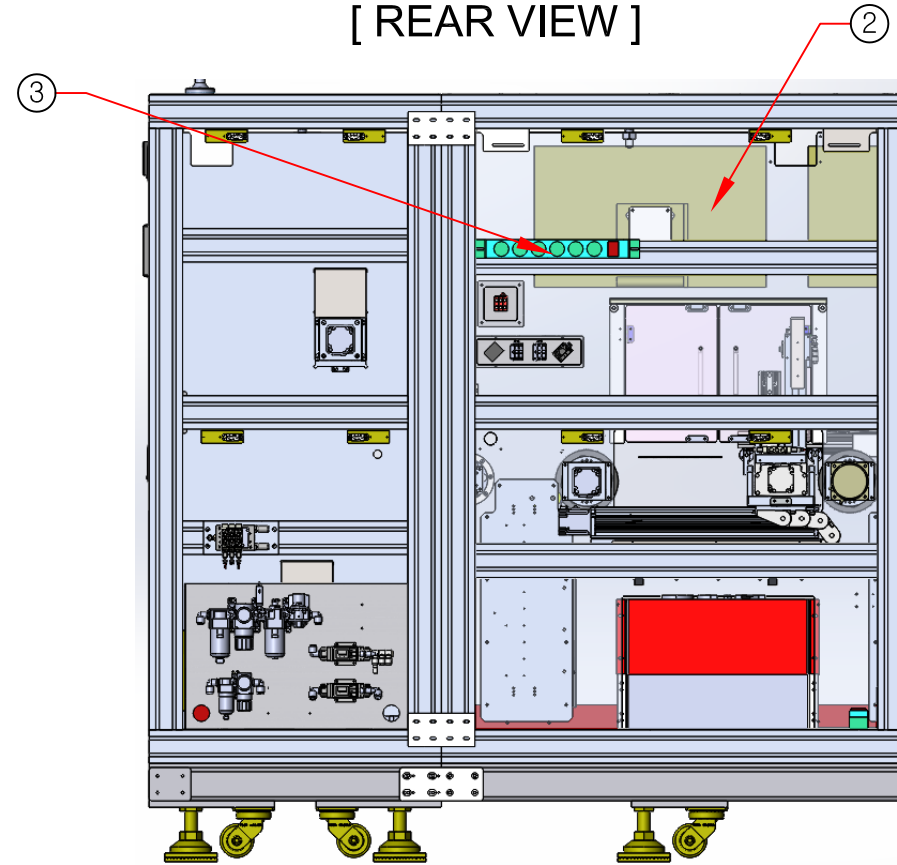
Designed	Modify	Approved	Material	Quantity	Treatment	Remarks
김관호 조득희		김관호	Machine name : SMART-IC AVI			
Date:		Unit : mm	Part name : VISION SYSTEM : 2nd FLOOR (TOP VIEW)			
Synapse imaging			Size A3	Drawing No.		Sheet No L25

MONITOR 1, 2 & 멀티탭 4

[FRONT VIEW]



[REAR VIEW]



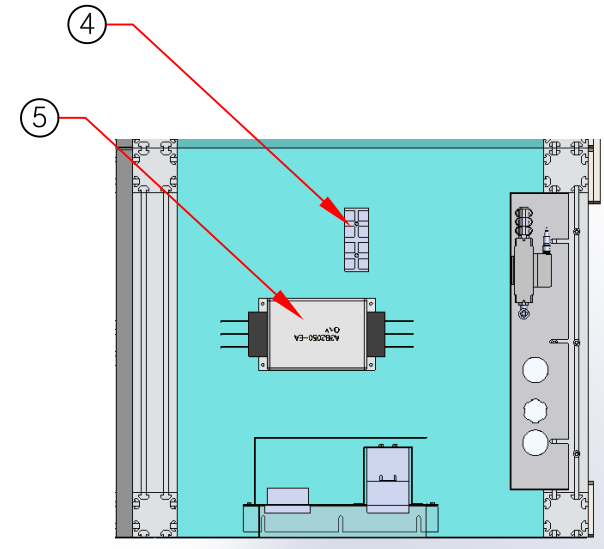
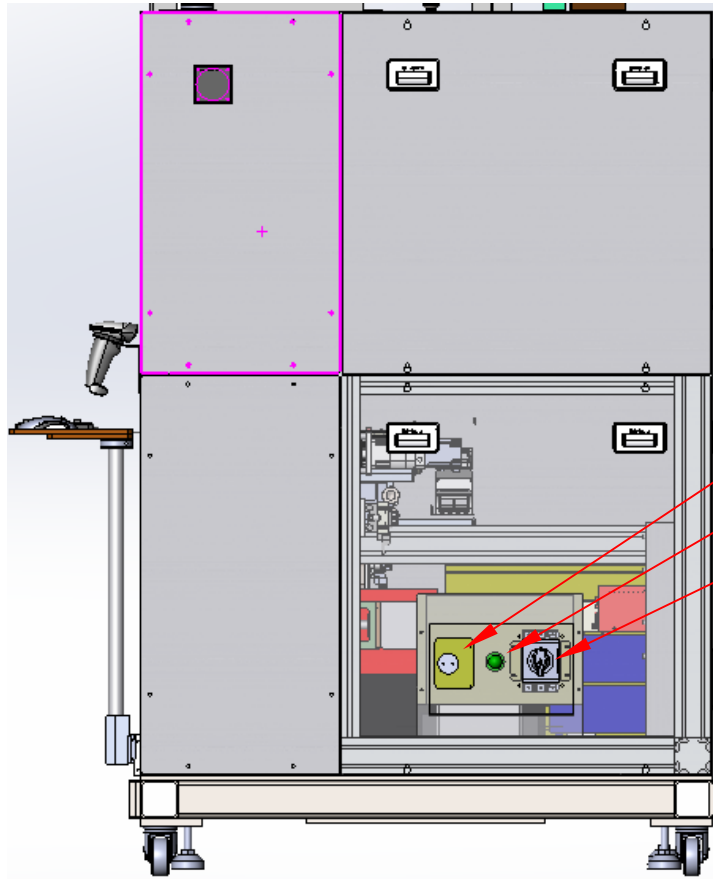
NO	NAME
①	MONITOR 1
②	MONITOR 2
③	L2-4S-5P / 4구, 5m, 알루미늄, 서버랙용

[작업 노트]

1. 후면부 프로파일에 멀티탭 4(4구 Type)를 설치한다. (C-BOX 1 Panel 설치 되면 가려져서 안보인다)
2. 멀티탭 4의 위치는 아답터 2개가 정상적으로 삽입 되도록 선정 한다.
3. 배선은 멀티탭 4의 Cable(5m)에 AC Plug (IEC320 C14)를 작업해서 UPS에 삽입한다.

Designed	Modify	Approved	Material	Quantity	Treatment	Remarks
김관호 조득희		김관호	Machine name : SMART-IC AVI			
Date: 16.08.25		Unit : mm	Part name : MONITOR 1, 2 & 멀티탭 4			
Synapse  imaging			Size A3	Drawing No.		Sheet No L31

UTILITY PANEL : RIGHT SIDE



* Main Noise Filter 및 주전원 단자대는 Utility Panel
후면 안쪽에 위치함.

NO	LABEL	MODEL	수량
①	ELCB (주전원 차단기)	EBN53C/50/100	1
	ELCB HANDLE	DH100	1
②	수전램프	L2RR-L1WL	1
③	SERVICE SOCKET	KDU-200-9LUA	1
		1x AC220V, 7A 콘센트	
		1x RJ-45 (MES I/F)	
④	TB 00	KTB1-06004 (4P, 60A)	1
	(주 전원 연결 단자대)		
⑤	NF 00	A3B2050-EA	1
	(Main Noise Filter)		

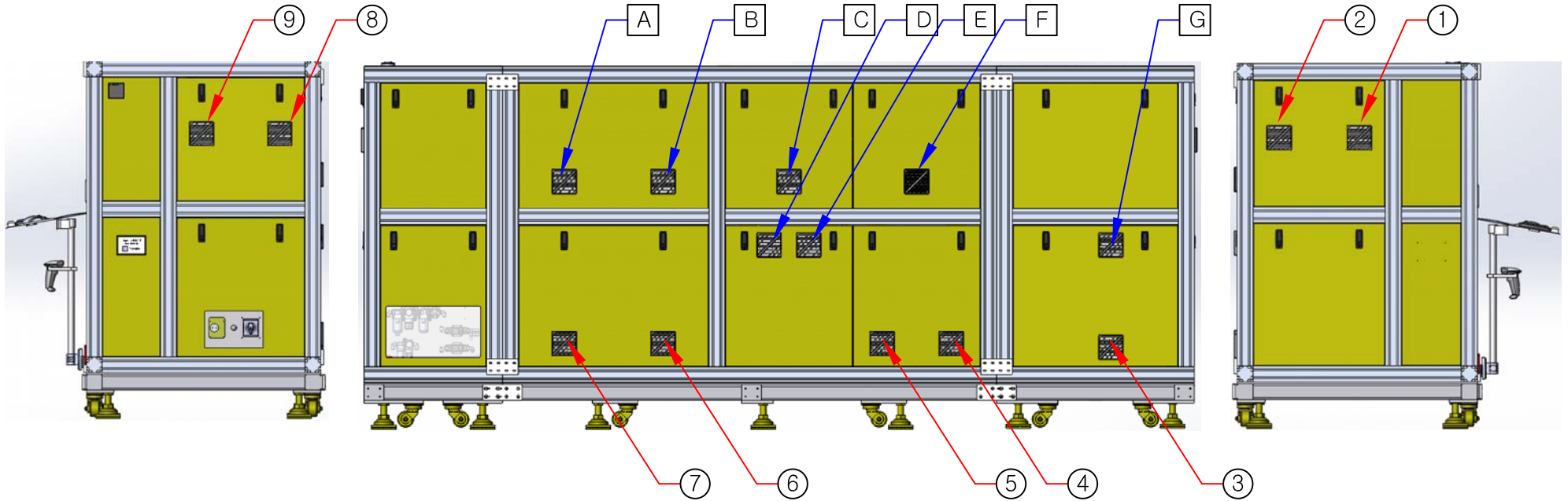
Designed	Modify	Approved	Material	Quantity	Treatment	Remarks
김관호 조득희		김관호	Machine name : SMART-IC AVI			
Date:		Unit : mm	Part name : UTILITY PANEL : RIGHT SIDE			
Synapse  imaging			Size A3	Drawing No.		Sheet No L32

COVER : COOLING FAN (REAR VIEW)

[RIGHT SIDE]

[REAR SIDE]

[LEFT SIDE]



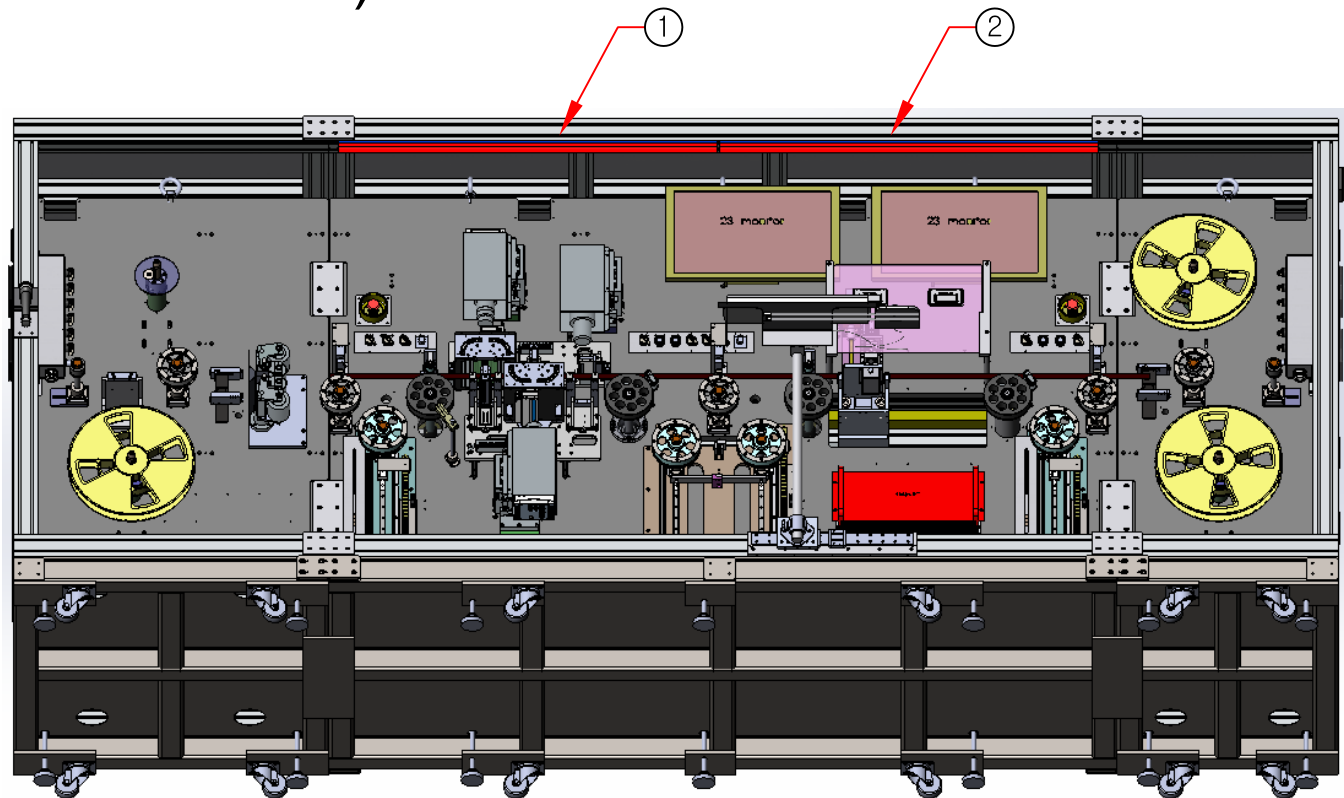
[작업 노트]

1. Cooling FAN 번호는 설비 전면을 기준으로 좌측면 상단 Cover FAN 부터 1번.
2. Cooling FAN의 Air Flow는 모두 흡기 방향으로 설치 한다.
3. FAN 이 설치 되지 않는 곳은 FAN 사각 Filter만 설치 한다. (A~G)

NO	LABEL	MODEL	수량
①	FAN 1	DP201A-2123HST.GN	1
②	FAN 2	DP201A-2123HST.GN	1
③	FAN 3	DP201A-2123HST.GN	1
④	FAN 4	DP201A-2123HST.GN	1
⑤	FAN 5	DP201A-2123HST.GN	1
⑥	FAN 6	DP201A-2123HST.GN	1
⑦	FAN 7	DP201A-2123HST.GN	1
⑧	FAN 8	DP201A-2123HST.GN	1
⑨	FAN 9	DP201A-2123HST.GN	1

Designed	Modify	Approved	Material	Quantity	Treatment	Remarks
김관호 조득희		김관호	Machine name : SMART-IC AVI			
Date: 16.08.25		Unit : mm	Part name : COVER : COOLING FAN (REAR VIEW)			
Synapse  imaging			Size A3	Drawing No.		Sheet No L41

LAMP 1, 2 (FRONT VIEW)



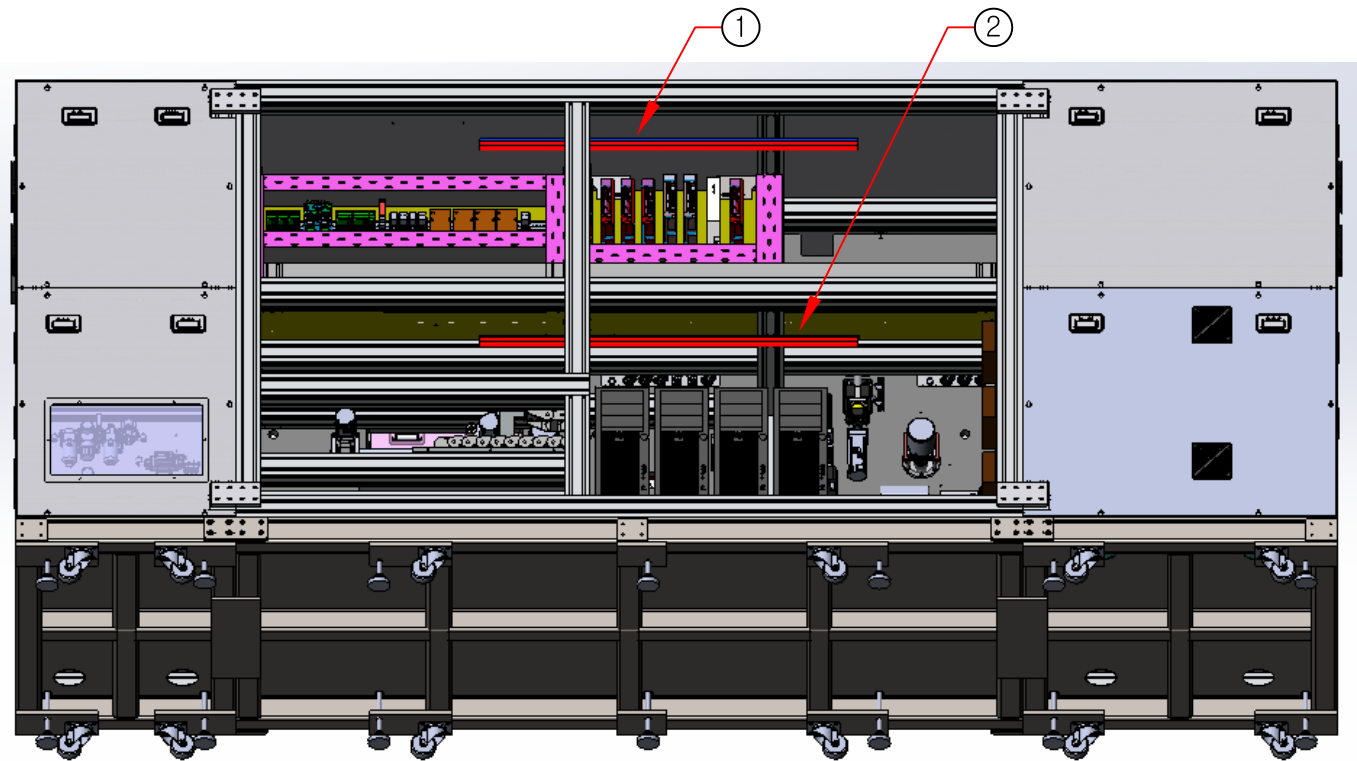
[작업 노트]

1. LAMP에 부착되어 있는 고정 클립을 사용하여 장착 한다.
2. LAMP 각도 조절이 설비 측으로 조절 될 수 있도록 장착 한다.
3. LAMP 1번과 2번을 연결 할 때는 직접 연결잭(부속품) 또는 연결 케이블(별매품) 을 사용한다.

NO	LABEL	MODEL	수량
①	LAMP 1	EZ LED WIDE 1200/18W/5700K/확산	1
②	LAMP 2	EZ LED WIDE 1200/18W/5700K/확산	1

Designed	Modify	Approved	Material	Quantity	Treatment	Remarks
김관호 조득희		김관호	Machine name : SMART-IC AVI			
Date:		Unit : mm	Part name : LAMP 1, 2 (FRONT VIEW)			
Synapse imaging			Size A3	Drawing No.		Sheet No L42

LAMP 3, 4 (REAR VIEW)



[작업 노트]

- 1. LAMP에 부착되어 있는 고정 클립을 사용하여 장착 한다.
- 2. LAMP의 장착 위치는 유지, 보수 작업시 어두워서 잘 보이지 않는 곳을 조명 할 수 있는 곳으로 선정 한다.
- 3. 노출 중간 스위치의 장착 위치는 사용자가 동작 시키기 편리한 위치를 고려하여 선정 한다

NO	LABEL	MODEL	수량
①	LAMP 3	EZ LED COMBI 1200/18W/5700K/확산	1
②	LAMP 4	EZ LED COMBI 1200/18W/5700K/확산	1

Designed	Modify	Approved	Material	Quantity	Treatment	Remarks
김관호 조득희		김관호	Machine name : SMART-IC AVI			
Date:		Unit : mm		Part name : LAMP 3, 4 (REAR VIEW)		
Synapse  imaging		Size A3		Drawing No.		Sheet No L43